DIOGO GOMES DE ARAÚJO

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Education

Master's **Sep 2025 – Present**
Computer Science *Universidade do Porto (FCUP)*
Focused on Distributed Systems, Programming Languages and Formal Verification.

Bachelor's **Sep 2022 – Jul 2025**
Computer Science and Engineering *Universidade da Beira Interior*
Final project presented at Inforum 2025 in Évora.

Projects

Purr Functional Programming Language | Haskell, Yacc, Lex, Type Inference, Lazy Evaluation

- Built a **lazy functional programming language** with ML-like syntax, including lexer, parser, compiler and interpreter.
- Implemented **Hindley-Damas-Milner type inference** with polymorphic functions and compile-time type checking.
- Implemented **lazy evaluation** based on graph-reduction techniques and developed a standard library of higher-order functions.

Garbage Collectors | C, OCaml, OCamlLex, Menhir

- Implemented three garbage collection algorithms for binary search trees: Mark & Sweep, Mark & Compact, and Copying Collection.
- Made a parser for a custom instruction set in OCaml and integrated it with C-based mutator and virtual-machine executables.

Fun, a Functional Language that Compiles to SECD | OCaml, Menhir, OCamlLex, λ-Calculus, SECD

- Translated an extended call-by-value λ-calculus into SECD instructions, implementing both the compiler and the stack-based abstract machine.
- Extended the language with pattern matching, recursive definitions, fixpoints and currying.
- Added a REPL and execution tracing to inspect compiled instructions and follow program evaluation.

Work Experience

INESC TEC **Jan 2026 – Apr 2026**
Researcher *Hybrid*

- Maintained **IoT gateway microservices** for communication with Distributed Energy Resources (DER).
- Deployed a **Zero Trust** setup (Teleport) giving secure remote access to every IoT device in the Power and Energy Systems lab.

STAR Junior Enterprise **May 2024 – Aug 2025**
Project Manager and Backend Developer *Remote*

- Developed **backend REST APIs** for mobile and web applications.
- Led project teams and worked directly with clients to deliver on requirements and on time.
- Screened candidates during recruitment.

Certifications

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch **Oct 2025**
Built a complete game in 48 hours with Rust and Raylib, including game mechanics, rendering and input handling.

Participation in Inforum 2025 | Inforum **Sep 2025**
Presented my Bachelor's final project to researchers, professionals and students.

Industrial Robotics Fundamentals | Universidade da Beira Interior **Jul 2025**

Volunteering | Missão País **Feb 2024**

Participation in RoboCup 2016 | RoboCup Federation **Jun 2016**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança **May 2016**

Technical Skills

Languages: Rust, OCaml, Haskell, C, Java, SQL
Distributed Systems & Networking: libp2p, QUIC, gRPC
Systems & Concurrency: UNIX, POSIX, OpenMP, STM, Tokio
Tools & Infrastructure: Git, Docker, Teleport, PostgreSQL, Raylib
Formal Methods & Testing: Quint, Rocq, QuickCheck
Web: HTML5, CSS, Dioxus, Actix Web, Bonsai, Dream
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge)